

Performing Social Closure: How College Board Market Devices Structure High School Recruiting Visits by Selective Colleges

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ABSTRACT

Sociology of education investigates consumer-facing intermediaries (e.g., college rankings), but pays less attention to business-facing intermediaries that classify students on behalf of schools/colleges. The market devices literature offers useful concepts. Market devices are calculative tools incorporated into products or economic transactions. Performativity is the idea that actors utilize market devices in ways that enact behavior consistent with theories of market behavior. College Board Geomarkets and the Market Segment model are market devices for higher education recruiting. Geomarkets carve metropolitan areas into smaller geographic units meant to define local recruiting markets. The Market Segment model predicts how student demand for a college varies by Geomarket as a function of social class. These market devices were the basis for the College Board Enrollment Planning Service, an early software-as-a-service product that helped colleges plan recruiting. Using data on recruiting visits to high schools made in 2017 by a sample of 42 colleges, we examine whether the relationship between recruiting visits and Geomarkets is consistent with theoretical expectations about performativity. [MODIFY ABSTRACT TO EMPHASIZE SOCIAL CLOSURE]

1 Introduction

Selective colleges are central sites of social closure. They confer access to scarce credentials and networks that facilitate entry into privileged occupations (Chetty et al. 2023), while enrolling student bodies that remain sharply stratified by class and race (Chetty et al. 2020a; Abramitzky et al. 2024). Weberian analyses emphasize the exclusionary strategies of status groups organized around a particular college, who redefine admissions criteria in ways that benefit the privileged group (Karabel 2005, 1984). These admissions criteria diffuse across selective colleges through field-level mechanisms of mimetic and normative isomorphism [CITE BROADER LITERATURE]. However, these explanations presume that social closure is enacted at the point of selection, overlooking how recruiting markets are constructed through shared infrastructures. If access to selective colleges is a mechanism of social closure structured along socioeconomic and racial lines [CITE HAMILTON], then a central question is how the recruiting markets from which applicants are drawn are delimited, commensurated, and rendered actionable.

In contemporary college recruiting, third-party “market devices” classify high schools and communities on behalf of colleges looking for students. *The Structure of College Choice* (Zemsky and Oedel 1983) developed Geomarkets and the Market Segment Model. Geomarkets carve metropolitan areas into smaller geographic units meant to define local recruiting territories. The Market Segment Model decomposed students into four market segments – local, in-state, regional, and national – based on the distance students were willing to travel for college. Zemsky and Oedel (1983) observed that student market segment was a function of class. Further, selective colleges primarily enrolled students from the “regional” and “national” segments, but the relative abundance of these students differs across Geomarkets. Over time, Geomarkets became an organizing principle for allocating admissions personnel to recruiting territories [CITE]. The Market Segment Model is the foundation of College Board’s popular Enrollment Planning Service (EPS) software, which encouraged colleges to identify recruiting territories by investigating which Geomarkets demonstrated strong student demand for peer colleges. We argue that Geomarkets and the Market Segment Model facilitate social closure by creating an infrastructure that encourages selective colleges to concentrate their recruitment efforts within affluent Geomarkets across the U.S. This study examines how Geomarkets and the Market Segment Model structure high school recruiting visits by a sample of 42 selective

colleges. [GPT says this paragraph still needs work].

We develop a conceptual framework that integrates scholarship on market devices (Callon 1998) and classification situations (Fourcade and Healy 2013). Market devices are tools that calculate and classify, which become incorporated into organizational goals, roles, and routines in ways that produce a menu of possible and likely actions. Geomarkets are market devices that become embedded in organizational routines and in other market devices, including Slate customer relationship management software, thereby defining the territories through which recruiting is organized. The Market Segment Model organizes student market segments – local, in-state, regional, and national – and colleges into a vertical hierarchy defined by class (Zemsky and Oedel 1983), whereby selective colleges primarily enroll “regional” and “national” students, who come from high-SES families, and are concentrated in wealthy Geomarkets. We argue that the patterns of high school visits by selective colleges perform the expectations of the Market Segment Model because Geomarkets and the logic of the Market Segment Model are embedded in the infrastructure that produces recruiting visits. This performance contributes to social closure because recruiting visits of selective private colleges and public research universities are concentrated in affluent, predominantly white communities.

We examine the relationship between College Board Geomarkets and high school recruiting visit patterns. We collected data about off-campus recruiting visits made in 2017 from a sample of 42 colleges, including 14 selective private research universities, 12 selective private liberal arts colleges, and 16 public research universities. We merge recruiting visit data with secondary data on high school and neighborhood characteristics. Analyses are guided by three research questions. First, to what extent are visit patterns consistent with recommendations from the Market Segment Model? To what extent do Geomarkets explain which high schools receive recruiting visits? How does Geomarket popularity – measured as number of visits per high school – mediate the relationship between school socioeconomic and racial/ethnic composition and the probability of receiving a visit?

For the selective private – and most public – institutions in our sample, the Market Segment Model recommends focusing on affluent Geomarkets located in regions where students are likely to consider a college like yours (Zemsky and Oedel 1983). Descriptive statistics show that recruiting visit patterns are more expansive geographically than enrollment patterns, but tend to be concentrated in adjacent regions except for highly selective institutions that enjoy strong national demand (e.g.,

Williams, UC-Berkeley). Visits by private colleges and out-of-state visits by public universities are highly concentrated in affluent Geomarkets. With respect to RQ2, fixed effects regressions indicate that Geomarkets explain substantial variation in recruiting visits even after controlling for a strong set of school- and neighborhood-level controls. With respect to RQ3, regression models find that schools with a low share of free/reduced lunch students benefit more from being located in a popular geomarket than schools with a high share of free/reduced lunch and schools with a modest share of Black/Hispanic students benefit more from being located in a popular Geomarket than schools with a high share of Black/Hispanic enrollment. This finding is consistent with the idea that the goal of classification situations is to create “good matches” between an ordinal hierarchy of products and an ordinal hierarchy of consumers (Fourcade and Healy 2017). Compared to a poor school, a wealthy school benefits more from being located in a wealthy Geomarket because this school is consistent with what college recruiters expect.

We make targeted contributions to the sociology of education and a broader contribution to the mechanisms of social closure. Scholarship on selective college access has focused on conflict over admissions criteria and which applicants benefit [CITE]. Scholarship on enrollment management shows that colleges devote substantial resources identifying and grooming potential applicants [CITE]. Research on both admissions and recruiting conceive center colleges as protagonists, ignoring business-facing intermediaries that classify students on behalf of colleges. Our research points to an upstream ecosystem of third-party vendors, products, and professional associations that collectively construct the infrastructure that colleges utilize to plan and implement their recruiting strategy. Geomarkets and the Market Segment Model were grafted on top of the history of racial and class segregation in America and became embedded within third-party products and college organizational structures. These market devices facilitate a tiered matching of colleges to places by encouraging selective private colleges to target high-income localities nationwide, while working class communities are left to less selective institutions. Although public research universities were engines of social mobility in the era of strong state funding, the desire to increase nonresident enrollment encourages public research universities to target the same set of affluent Geomarkets coveted by selective private institutions. Therefore, Geomarkets and the Market Segment Model contribute to social closure by engendering an infrastructure that matches schools and communities

to colleges based on socioeconomic privilege.

The manuscript is organized as follows. First, we review scholarship on college access and enrollment management. Second, we describe the case of Geomarkets and the Market Segment Model. Third, we develop a theoretical framework based on market devices and classification situations. Fourth, we describe data and methods. Fifth, we present results. Finally, we discuss implications for scholarship. Scholarship at the intersection of quantification and education has considered consumer-facing intermediaries (e.g., college rankings) that discipline consumers and, in turn, businesses. However, intermediaries that classify consumers on behalf of businesses are more pervasive and more sophisticated. In the education industry, these intermediaries match ordinally arranged consumers to ordinally arranged products. We argue that sociology of education can develop novel insights by shifting the object of inquiry from students and direct providers education (schools, and colleges) to the market devices that classify students for direct providers. [something about undermatch?]

2 Literature Review: College Access in the Enrollment Economy

Social closure, associated with Max Weber [CITE], is the process by which advantaged groups maintain their privilege by restricting access to resources and opportunities that confer economic and social privilege. Weber conceived of social closure as competition between status groups, defined of “associational communities comprised of all who share a sense of status equality” (Karabel 1984, 3) based on membership of a characteristic (e.g., racial, ethnic, religious, social) that may be the basis of exclusion. Following the transition to rational-legal authority defined by universalistic bureaucracies, organizations that confer economic and social privileges become the site of contestation between dominant groups seeking exclusionary closure and marginalized groups seeking usurpationary closure. Colleges are the exemplar site of contestation (Collins 1979). Weber states,

The development of the diploma from universities ...make for the formation of a privileged stratum in bureaus and in offices. Such certificates support their holders’...claims to monopolize socially and economically advantageous positions. When we hear from all sides the demand for an introduction of regular curricula and special examinations, the reason behind it is, of course, not a suddenly awakened ‘thirst for education’ but

the desire for restricting the supply for these positions and their monopolization by the owners of educational certificates...As the acquisition of the educational certificate requires considerable expense and a period of waiting for full remuneration, this striving means a setback for talent (charisma) in favor of property.

In the U.S., selective colleges confer access to privileged social and economic positions. As college access expands to previously excluded populations, scholarship in sociology and economics document growth in the benefits of graduating from a selective institution (Chetty et al. 2023; Gerber and Cheung 2008), and growth in competition for access to selective institutions [CITE]. Based on data from the IRS, the “Mobility Report Cards” (Chetty et al. 2020b) measures both parental household income and student earnings post-graduation at the college level for students who were born in 1991 and were freshman around 2009. Although low-income students students who graduate from selective colleges have strong labor market outcomes (Chetty et al. 2020a), access to selective institutions is dominated by wealthy households. For example, at Colorado College, one of the colleges in our analysis sample, median family income was about \$379,000 (2026 CPI), with just 2% of freshman coming from the bottom income quintile, 78% of freshman coming from the top quintile, 54% of freshman from the top 5%, and 24% of freshman from the top 1% (Chetty et al. 2020b).

Several literatures in the sociology of education have analyzed the admissions criteria utilized by selective colleges and which applicants these criteria benefit. The Weberian status group social closure tradition is articulated by the historical sociology of (Karabel 2005, 1984). Leaders of Ivy League institutions were themselves members of the elite Protestant status group and had an interest in blocking Jewish enrollment growth by developing “character” admissions criteria associated with distinctly upper-class Protestant activities and traits. [TRANSITION?] A robust literature analyzes the rise and fall of affirmative action (Hirschman et al. 2016; Grodsky 2007; Warikoo 2025; Karen 1991). Another literature ties the growing importance of SAT/ACT scores in selective college admissions to the rise of college rankings systems that reward colleges for enrolling high-scoring students (Grodsky et al. 2008; Alon and Tienda 2007; Espeland and Sauder 2016). Recent scholarship examines the implementation of “test optional” policies (Hsu and Sharkey 2025) and holistic admissions processes that evaluate applicants with respect to the opportunities available in their local context [CITE BASTEDO]. Although these studies offer rich accounts of how colleges

evaluate applicants, the scholarly preoccupation with selective admissions obscures the basic fact that overwhelming majority of colleges – including most selective colleges – cannot thrive by passively receiving applications.

The “enrollment economy” refers to reliance on student enrollment for organizational stability. The concept was introduced by Clark (1956) [p. 332], in his analysis of California adult education institutions: “The most important pressures on these schools in their day-to-day administration arise from the *enrollment economy* ...Financial support from the state is figured by the hours of attendance logged the previous year ...and a major slump in attendance constitutes a serious threat to organizational welfare.” (p. 332). At the opposite end of the prestige spectrum, Winston (1999, 17) describes dependence on students by selective private colleges: “the technology of producing much of what is sold in higher education is unusual in that colleges can buy important inputs to their production only from the customers who buy their products; that is, higher education uses a customer-input technology. Colleges depend on enrolled students for tuition revenue and on alumni for donation revenue. They depend on enrolled students for peer learning effects. Furthermore, the characteristics of enrolled students – for example, average SAT scores – substantially determine their national ranking, which then determines student demand for the subsequent cohort of prospects [CITE] and also affects research funding and donation revenue [CITE].

Organizational stability in the enrollment economy is the responsibility of enrollment management (EM). As a profession, EM integrates techniques from marketing and economics in order to “influence the characteristics and the size of enrolled student bodies” (Hossler and Bean 1990, xiv). As an administrative structure, the “office of enrollment management” typically controls and coordinate the functional activities of admissions, financial aid, and recruiting (Kraatz et al. 2010; Hossler and Bontrager 2014).¹ The administrative structure of EM offers one means of categorizing scholarship about the supply-side of college access. Accross disciplines, the majority of scholarship focuses on admissions (e.g., Hirschman et al. 2016; Taylor et al. 2024; Posselt 2016; Killgore 2009) or on financial aid (e.g., Hurwitz 2012; Leeds and DesJardins 2015). These literatures offers less insight into how colleges identify and cultivate prospective applicants before admissions evaluation occurs.

¹Kraatz et al. (2010) exemplifies a literature at the intersection of EM and organizational behavior that shows how concerns about the enrollment economy motivate changes in organizational behavior, such as diversifying degree programs or changing the name of the organization (Brint and Karabel 1989; Kraatz and Zajac 1996; Jaquette 2013)

Therefore, a growing literature in sociology analyzes the earlier “recruiting” stages of acquiring “leads,” and soliciting “inquiries” and applications [CITE]. Several monographs analyze recruiting as part of broader studies on enrollment management (Stevens 2007), high school to college transitions (Holland 2019), elite reproduction (Khan 2011), or political economy (Cottom 2017). From the student perspective, Holland (2019) finds that underrepresented students were drawn to colleges that made them feel wanted, often attending institutions with lower graduation rates and requiring larger loans than other college options. Cottom (2017) shows that for-profit colleges found a niche in Black and Latinx communities because traditional colleges ignored these communities.

Stevens (2007) provides an ethnography of enrollment management at a selective private liberal arts college. Being tuition-dependent and sensitive about its position in the rankings, The College focuses on enrollment goals related to academic profile and revenue generation. Stevens (2007) highlights the relational function of off-campus recruiting visits: “the College’s reputation and the quality of its applicant pool are dependent upon its connections with high schools nationwide” (p. 54). During the autumn “travel season,” admissions officers visit selected high schools across the country “to spread word of the institution and maintain relationships with guidance counselors” (p. 53-54). Ruffalo Noel Levitz (2020) reported that “calling high school counselors after visits” was an effective and widely-used outreach strategy for both private universities and public universities. Thus, relationship maintenance remains a high-touch process even in the digital age. The College tended to visit the same “feeder” schools year after year because recruiting depends on long-term relationships. Visited schools tend to be affluent schools – in particular, private schools – that possess the resources to host a visit and enroll high-achieving students who can afford tuition.

Whereas scholars in the sociology of education have investigated recruiting as part of a broader analysis, AUTHORS (2021; 2022; 2024) have attempted to develop a literature that makes college recruiting behavior the object of empirical analysis. Salazar et al. (2021) argue that off-campus recruiting visits are a credible indicator of college enrollment priorities.² [INSERT SENTENCES SAYING HOW MUCH IS SPENT ON OFF-CAMPUS RECRUITING AND SCHOLARSHIP ON THE EFFECTS OF OFF-CAMPUS RECRUITING]. Salazar et al. (2021) analyzed recruiting visits

²Sociological theories of organizational behavior argue that organizations expend substantial resources on goals they care about while symbolically adopting – highly visible, ceremonial behavior with few resources expending – goals demanded by stakeholders in the broader institutional environment [CITE].

made by public research universities. The authors found that 12 of the 15 universities made more visits to out-of-state schools than in-state schools. These out-of-state visits focused on affluent, predominantly white public high schools. Set against scholarship about nonresident enrollment growth as a response to declining state support (Jaquette and Curs 2015; Jaquette et al. 2016), these results suggest that many public research universities prioritize enrollment from wealthy out-of-state students.

Jaquette et al. (2024) conduct a social network analysis of recruiting visits to private high schools by a sample of public research universities and selective private research universities. A visit indicates a social relationship between a college and a high school, suggesting that the college wants to enroll some students from the high school and the high school wants to send some students to the college. Both public and private universities made a disproportionately large number of visits to private schools relative to public schools. Analyses revealed high overlap in the recruiting networks of public and private universities. Salazar (2022) conducted geospatial analyses of out-of-state recruiting visits to Los Angeles and Dallas by four public research universities. She found that universities engage in “recruitment redlining – the circuitous avoidance of predominantly Black and Latinx communities along recruiting visit paths” (p. 585).

Upon review, scholarship locates inequality in college access in the behavior of households, schools, and colleges [CITE]. Scholarship on the supply-side of college access – selective admissions, EM, recruiting – argues that students are sorted by the actions of colleges, which are conditioned by macro-institutions in the organizational field [CITE]. This focus on college behavior understates the role of third-parties that mediate between consumers and sellers. Sociology of education has investigated intermediaries that “occupy the interface between consumers and producers” (Sharkey et al. 2023, 1), including scholarship on college rankings (Espeland and Sauder 2016) and school rankings (McArthur and Reeves 2022; Thompson 2024; Hasan and Kumar 2019). To our knowledge, intermediaries that classify consumers on behalf of sellers have rarely been the object of empirical research (but see Hamilton et al. 2024). Although consumer-facing intermediaries are salient to media outlets and researchers, intermediaries that classify consumers for businesses are more sophisticated, pervasive (McKelvey 2022; Muniesa et al. 2007; Cochoy 2007; Fourcade and Healy 2017), and may be highly salient to professionals responsible for recruiting customers. Scholarship

on college access has ignored the infrastructural mechanism of third-party enrollment management tools that classify students and localities on behalf of colleges. These tools – which are built atop the foundation of segregation in American communities – facilitate a tiered matching of colleges to places by encouraging selective private colleges and public research universities target high-income localities nationwide, while encouraging less selective institutions to target nearby working- and middle-class communities.

Geomarkets were conceived in service of a class-based model of college choice (Zemsky and Oedel 1983). Subsequently, Geomarkets became the basis of College Board Enrollment Planning Service (EPS) software that helped colleges plan recruiting efforts and an organizing principle allocating admissions personnel to recruiting territories. We conceptualize these tools as “market devices” (Callon 1998) that create “classification situations” (Fourcade and Healy 2013), whereby an ordinal hierarchy of students is matched to an ordinal hierarchy of postsecondary institutions. Performativity is the idea that market devices are used in ways that create the behaviors they predict (MacKenzie and Millo 2003). Empirically, we show that Geomarkets explain substantial variation of high school visits by a sample of selective private colleges and public research universities, even after controlling for a strong set of school- and neighborhood-level covariates. Finally, we contribute to a critical sociology of the College Board. Existing scholarship focuses on College Board’s consumer-facing testing products [CITE], mirroring the blind-spot observed in scholarship on intermediaries. This manuscript shows that College Board testing products are the basis for business-facing products that classify prospective students on behalf of colleges.

3 The Case: Geomarkets and the Market Segment Model

Geomarkets were conceived in the late 1970s, when colleges were concerned about declining demographics. Zemsky and Oedel (1983) [p. ix] write that demographer “Stephen Dresch (1975) had already frightened everyone who would listen with his projection of 40-percent declines in college enrollments by the close of the 1980s.” Elite colleges responded by leaning on the associational configuration (Stevens and Gebre-Medhin 2016) to solve problems common to the group. The project began when UPenn president, Martin Meyerson, tasked UPenn professor Robert Zemsky, to figure out, “‘Who thinks about Penn?’” and “‘What other institutions do they think about when

they think about us?’ ” (Zemsky and Oedel 1983, x):

“In pursuit of these questions, we began working in 1978 with the Market Research Committee of the Consortium on Financing Higher Education (COFHE). The Consortium, a group of thirty selective private institutions, had in its brief history developed a remarkable tradition of sharing data...For nearly a year, the Consortium’s Market Research Committee provided focus to our efforts. Much of this project’s framework – the Market Segment Model and its derivation from admissions officers’ folklore was shaped and then reshaped in our discussions with members of the committee.”

Zemsky and Oedel (1983) conceived of the Market Segment Model as an effort to “capture and quantify” (p. 11) the knowledge of admissions officers: “We believe that the intuitions of admissions officers actually comprise a remarkably systematic body of knowledge about the college selection process...Our research is based upon listening carefully to what admissions officers have to say (pp. 9-10). The”initial [analytic] task” was to define geographic boundaries “in a manner consistent with admissions officers’ intuitive understanding of student pools” (Zemsky and Oedel 1983, 4). Quoting an admissions officer, Zemsky and Oedel (1983) [p. 11] write, “‘There are only three kinds of college-bound students: those who want to live at home, those who want to live on campus but bring their laundry home, and those who want to go far enough from home that Mom and Dad can’t visit without calling first.’ ” As such, Zemsky and Oedel (1983, 11) created “three types of boundaries – region, state, and community.” The initial regions in the project were New England, Middle States, and the South. Next, they “divided each state into as few as two and as many as thirty community-based enrollment markets or pools, for a total of 143 separate markets” (p. 11). These enrollment markets, later called Geomarkets, were intended to be consistent with the conception of a catchment market from the perspective of admissions counselors. Zemsky and Oedel (1983, 11–12) described the creation of Geomarket borders only briefly:

In many cases, the market boundaries match formal political and educational divisions, reflecting natural channels of communication. Each major metropolitan area is composed of several markets, usually corresponding to the inner city, a first ring of suburbs, and an outer ring of suburbs. In more sparsely populated areas, communities are sometimes combined in order to make the analysis meaningful.

Over time, the project goal morphed into predicting student demand for any college from any locality. However, “to gain a truly comprehensive view of the collegiate enrollment market, we needed a database that described most institutions and most students” (Zemsky and Oedel 1983, x). The group approached College Board. “Coincidentally, the Board was reviewing its own efforts to help colleges estimate their enrollment potential” (Zemsky and Oedel 1983, x). College Board provided score-sending data for all 1980 SAT test-takers. For a given student, each college that receives a score from the student can be defined as “local” (college located in the the same Geomarket as the student), “in-state” (same state but different Geomarket), “regional” (same region but different state), or “national” (different region). Zemsky and Oedel (1983) classified each SAT test-taker into one of four student *market segments*. A student is categorized in the “local” market segment if they submit more SAT scores to local institutions than they do to in-state, regional, or national institutions. An “in-state” student submits more SAT scores to in-state institutions than they do to local, regional, or national institutions, etc. The Market Segment Model “was nothing more than a set of simple rules for disaggregating high school seniors into similar groups” (Zemsky and Oedel 1983, 4). “The model worked because students, once so disaggregated, appeared to behave in remarkably consistent ways.”

Zemsky and Oedel (1983, 3) investigates which student characteristics predict whether a student belongs to the local, in-state, regional, or national market segment. The analyses identify four variables – educational aspirations, parental education, scholastic aptitude, and family income – that predict score-sending behavior, both individually and in combination. The resulting thesis is that student demand for college is a function of social origin. These four variables “reflect the basic social patterns of the nation; it would have been surprising if these were not the four social variables that best explained the patterns of college choice” (Zemsky and Oedel 1983, 33). The analyses conclude (p. 42) that “our research has simply demonstrated what everyone has always known: communities with high levels of family income and parental education are also communities in which students have higher than average SATs and more far-reaching aspirations.”

However, Geomarkets differ in the relative abundance of students with particular socioeconomic characteristics. As college leaders sought new sources of patronage, the Market Segment Model yielded practical implications about which communities to target. Zemsky and Oedel (1983,

44) recommend that colleges target Geomarkets with desirable compositions of socioeconomic characteristics in order to reach students from desired student market segments:

On occasion, senior spokespersons for the profession worry that students outside the main [Geo]market areas remain forgotten and hence, unchallenged. Inevitably, the increasing competition for students, the expense of travel and mailings, and internal political constraints compel institutions to concentrate their efforts where they will do the most good. The result is a natural reinforcing of the basic socioeconomic patterns that gave shape in the first place to the structure of college choice.

Zemsky and Oedel (1983) developed two analytic outputs, (1) the Market Segment Profile and (2) the Institutional Profile. The Market Segment profile showed – separately by Geomarket – the number of students from each market segment (local, in-state, regional, national), their socioeconomic characteristics, and average test scores. The Institutional Profile showed – separately by Geomarket and institution (your own or a competitor) – how many students from each market segment sent scores to that institution.

In 1984, College Board transformed the Market Segment Model and associated outputs into the Enrollment Planning Service (EPS), an early software-as-service platform sold to colleges that was based on Zemsky and Oedel (1983). For any Geomarket, colleges could obtain the Market Segment Report and the Institutional Profile for themselves or a competitor. Based on background conversations with EM professionals, EPS software also provided information about the score-sending behavior of individual high schools within each Geomarket. Typical College Board (2005) marketing material describes EPS as, “the marketing software that pinpoints the schools and Geomarkets where your best prospects are most likely to be found. With the click of a mouse, EPS provides you with comprehensive reports on your markets, your position in those markets, and your competition.” Following Zemsky and Oedel (1983), the EPS platform encouraged users to, first, identify promising Geomarkets and, second, to identify desirable high schools for recruiting visits. To find new markets, EPS encouraged colleges to select peer colleges and identify Geomarkets where a large number of students sent scores to those colleges. Market research suggests that EPS was highly salient. Noel-Levitz (1998) reports that in 1995, 37% of 4-year publics and 49% of 4-year privates used EPS, while 41% of 4-year publics and 16% of 4-year privates used ACT’s market analysis service product.

4 Theory: Market Devices and Classification Situations

The market devices literature examines how the components of markets are assembled and enacted, and how these processes shape behavior (Callon 1998; Muniesa et al. 2007; MacKenzie and Millo 2003). Market devices are discreet calculative tools (e.g., algorithms, metrics, graphs, rankings models, spreadsheets) that are incorporated into products or economic transactions. Muniesa et al. (2007, 2) define market devices as a “way of referring to the material and discursive assemblages that intervene in the construction of markets.” Here, material refers to tangible tools and technologies, like spreadsheets, algorithms or software, while discursive refers to representational elements, such as categorical, quantitative, and ordinal variables. Assemblages refers to a bundle of heterogeneous components that work together, suggesting that market devices consist of and build upon other market devices. Geomarkets are a market device – based on the SAT – that helps enrollment managers define recruiting territories and make decisions about which high schools to recruit from. Market devices are non-neutral, actively shaping outcomes. Therefore, markets do not exist a priori waiting to be observed; they are constructed by market devices. From the market devices perspective, the transformation of American higher education “from a collection of local autarkies to a nationally integrated market” (Hoxby 1997, 1) can be (partially) explained as the consequence of enrollment managers using Geomarkets to achieve enrollment goals.

Theories (e.g., pricing theory, status attainment theory), by themselves, are not market devices. They are abstract ideas that can be applied to specific contexts. For example, homophily is the idea that people who share similar characteristics are more likely to connect (McPherson et al. 2001). Theories become part of market devices when components of the theory are embedded within the device. Zemsky and Oedel (1983) argue that homophily is the organizing principle of competition between colleges; similar colleges compete with one another for similar students. An analytic output from Zemsky and Oedel (1983) is the Institutional Profile, which provided information about students from a particular Geomarket who sent SAT scores to a particular institution. This functionality was subsequently embedded in EPS software, which encouraged colleges to investigate which Geomarkets have high demand for peer colleges

In the market devices framework, neither market devices nor actors have agency in isolation from

one another. Whereas market devices are tools, “agencements” are “socio-technical configurations” (Çalışkan and Callon 2010) of market devices, people, and organizational arrangements (goals, roles, routines, incentives) designed to produce a menu of possible actions and to make some actions more likely than others. Agencement orients research to the intersection of what the market devices is capable of doing and the organizational context that is utilizing the device – in concert with other devices – to achieve organizational goals. Geomarkets are an input for enrollment management offices charged with realizing college enrollment goals. Geomarkets can be utilized to define recruiting territories and assign admissions officers to territories. EPS can be used to investigate which Geomarkets exhibit strong demand for the college, for peer institutions, and to plan which high schools in a Geomarket will be visited by which admissions officer.

Performativity is the idea that theories of market behavior do not merely describe behavior, rather market devices are constructed and acted upon in ways they bring about behavior consistent with the theory (Muniesa et al. 2007; MacKenzie and Millo 2003). For example, MacKenzie and Millo (2003) provide a historical sociology of the Chicago Board Options Exchange, which opened in 1973. An option is a contract that gives the holder the right to buy or sell an asset (e.g., a stock) at a given price. Economists Fischer Black and Myron Scholes developed the Black-Scholes model – a calculative market device – to predict the price of options Black and Scholes (1973), eventually winning the Nobel Prize in Economics. Initially, the Black-Scholes algorithm did not accurately predict the price of options. Black-Scholes model predictions became more accurate over time because options traders began embedding the logic and components of the model into spreadsheets/software. That is, the model’s predictions became more accurate because the model produced the prices it predicted (MacKenzie and Millo 2003)! Bastedo and Bowman (2010) provide an example of performativity for the case of US News College Rankings. The reputational assessment of senior administrators was a substantial component of the rankings algorithm. Bastedo and Bowman (2010) show that administrators based their assessment scores on the prior overall rankings. Therefore, the rankings algorithm disciplines administrators to provide assessments that are consistent with the prior rankings.

Although canonical contributions to the market devices literature did not focus on structural inequality – race, gender, class, Hirschman and Garbes (2019) argue that market devices literature

can offer novel insights about the mechanisms of structural inequality. Analysis of the Houston residential real estate market by Korver-Glenn (2018, 2022) shows how market devices can be a mechanism of racial inequality. Every home listing is assigned to a Houston Association of Realtors (HAR) “market area” – a market device akin to Geomarkets – based on the address of the property. In 70 of the 98 HAR market areas, one racial group comprised at least 50% of residents. Korver-Glenn (2022, 641) observed that “50 of these 70 market areas were majority-white – a disproportionately high number given that only 31 percent of the population within the encompassing area is white.” On average, majority-white market areas contained about 20,000 residents, while majority-Black and majority-Hispanic market areas, respectively, contained 61,000 and 88,000 residents. Majority-white market areas also exhibited greater socioeconomic homogeneity. Because appraisals are based on recent comparable sales within the same market area (agencement), homes in high-income white neighborhoods received higher appraisal values than homes in high-income Black or Hispanic neighborhoods. College Board Geomarkets often exhibit similar differences in granularity. For example, the affluent, predominantly White “IL 9 - North Shore” Geomarket, had a 2020 population of about 161,000 while “IL 12 - South and Southwest Suburbs” Geomarket had a 2020 population of about 1.2 million.

The concept commensuration (Fourcade 2011; Espeland and Stevens 1998) helps us gain deeper insight about the agencement of market devices created by College Board. Commensuration is “the expression or measurement – of characteristics normally represented by different units – according to a common metric” (Espeland and Stevens 1998, 315). Commensuration is a process that requires work, akin to institutionalization. Once achieved, commensuration reduces the cognitive effort of decision-making. It is “a way to reduce and simplify disparate information into numbers that can easily be compared. This transformation allows people to quickly grasp, represent, and compare differences” (p. 316). For example, college rankings enable prospective students to compare colleges based on a single metric. Therefore, commensuration “can be understood as a system for discarding information and organizing what remains into new forms” (Espeland and Stevens 1998, 317). McArthur and Reeves (2022) analysis of UK school “league tables” – a commensurating market device that facilitated comparing schools on the basis of test scores – shows how commensuration can be a mechanism of social reproduction; professional/managerial households were more aware of these

consumer-facing metrics and had resources to respond by moving to more expensive neighborhoods, near higher performing schools.

College Board Geomarkets are a commensurating market device that was built on the SAT and became an input into subsequent market devices. Drawing from Hoxby (2009), the SAT is a commensurating market device that enabled colleges to compare students from different schools, causing a “dramatic fall” in the cost of “colleges’ information about students” (p. 102). The Market Segment Model is a theory of student demand that categorizes students into four market segments – local, in-state, regional, and national – based on SAT score-sending behavior. Geomarkets are a market device that defines geographic localities and enables colleges to commensurate these localities based on the number of students from each market segment. Whereas UK league tables allowed parents to compare geographically disparate schools based on test scores (McArthur and Reeves 2022), Geomarkets and the Market Segment Model enables colleges to compare geographically proximate and geographically disparate Geomarkets based on the number of students from each market segment. Each student market segment is associated with particular sets of institutions. Zemsky and Oedel (1983) states that selective colleges primarily draw students from the regional and national market segments. The key to finding new student markets, then, is identifying Geomarkets with large numbers of regional and national students. Geomarkets are then incorporated into the Enrollment Planning Service product, enabling colleges to decide which Geomarkets to recruit from and which high schools to visit (Takamiya 2005).

Classification situations. Scholarship on classification situations (Fourcade and Healy 2013, 2024, 2017; Fourcade 2016) helps conceptualize how Geomarkets facilitate social and racial stratification in college access. Traditionally, sociology defined social class from the production side of the economy (e.g., worker vs owner; clerical, manager, professional). By contrast, Fourcade and Healy (2013) argue that processes of ordinalization “endogenous to the market” (p. 560) define class through effects on the *consumption* side of the economy.

Classification situations are ordinal interventions whereby organizations “sort and slot people into new types of categories with different economic rewards or punishments attached to them” (Fourcade and Healy 2013, 561). The canonical example is credit scores (Poon 2007), which enable third-party intermediaries on supply side to “slice consumers into behaviorally-defined risk groups, and price

offerings to them accordingly. On the demand side, consumers find themselves ...fitting into these categories – which, by design, are not constructed from standard demographic classifications such as race” (Fourcade and Healy 2013, 560) but are highly correlated with social categories. The goal of classification situations is “good matches,” whereby an ordinal hierarchy of products match to an ordinal hierarchy of people. Businesses “seek good matches when they want to link a wealthy customer with a quality credit card, or a heavy drinker with a bad insurance plan” (Fourcade and Healy 2017, 17). Whereas economics tends to view such matches favorably – for example, Hoxby (2009, 106) writes the “efficient outcome allocates students to colleges based on students’ ability to benefit from the ...magnitude of the human capital investment that the college offers” – Sociology tends to highlight the predatory, extractive inclusion of formerly excluded consumers (e.g., for-profit colleges) (Cottom 2020).

Geomarkets and the Market Segment Model function together as an integrated market device that matches a socioeconomic hierarchy of places to a socioeconomic hierarchy of colleges.³ Zemsky and Oedel (1983) defined student market segments in ordinal terms – local, in-state, regional, and national – and found that student market segment was primarily a function of social class, particularly parental income and parental education. Zemsky and Oedel (1983) also conceived of colleges as an ordinal hierarchy. As in Fourcade and Healy (2017), good matches are the key to realizing enrollment goals. In turn, Zemsky and Oedel (1983) argue that good matches are defined by homophily, the idea that actors who share common characteristics are likely to form connections with one another (McPherson et al. 2001; Chun 2021). Zemsky and Oedel (1983) conduct a network analysis whereby two colleges are defined as competitors if a large number of students submit SAT scores to both colleges, Zemsky and Oedel (1983) “draw[ing] a fundamental conclusion about the structure of college choice: collegiate competition occurs principally between like institutions” (p. 46). Subsequent analyses investigate the tuition price and the socioeconomic composition of institutions in competition with one another. The takeaway is that like-colleges compete for like-students as defined by social class, with private selective colleges competing directly for students, charging the highest prices, and enrolling students with the highest socioeconomic

³Whereas Fourcade and Healy (2024) analyze devices that classify individuals, the Market Segment Model harkens back to the earlier era of geodemography, in which market research classified places for merchants based on the attributes of people who lived there (Leyshon and Thrift 1999; McKelvey 2022)

status (p. 72):

Students describe themselves socially simply by telling us the colleges and universities in which they are interested. The layering of collegiate competition is primarily a socioeconomic layering. The hierarchical structure of collegiate competition largely reflects the stratified social and economic dimensions of the communities from which colleges draw their students. Competition among colleges, as admissions officers have told us for so long, is in fact, a matter of keeping company with one’s peers.

4.1 Research questions and expected results

Our overarching research question is, what is the relationship between college Board Geomarkets and high school recruiting visits of selective private colleges and public research universities? Colleges engage in enrollment management behaviors designed to realize enrollment goals. Whereas Geomarkets, the Market Segment Model, and EPS are market devices and/or components of market devices, the patterns of recruiting visits from colleges to high schools describe the performativity consequences of colleges using these market devices to realize enrollment goals.

We sketch the enrollment economy for our analysis sample, consisting of 12 selective private liberal arts colleges, 14 selective private research universities, and 16 public research universities. Selective liberal arts colleges hews most closely to the characterization of Winston (1999); their position in the zero-sum game for prestige substantially depends on the high school achievement of enrolled students. Primary revenue sources are tuition and donations from wealthy alumni. They primarily enroll rich students with a modicum of students from middle- and lower-class backgrounds (Chetty et al. 2020a). Only a handful of colleges (e.g., Williams, Swarthmore) enjoy strong demand from students who are both affluent and high achieving. Outside the top ten or so, the priority becomes “affluent families [who] could absorb the College’s full tuition and board costs with little, if any, financial aid” (Stevens 2007, 73). Selective private research universities face a similar enrollment economy as liberal arts colleges, but are less dependent on students to the extent that they derive financial and reputational resources from faculty research. Following World War II, public research universities were less beholden to the enrollment economy due to federal research funding and stable state appropriations (Stevens and Gebre-Medhin 2016). State appropriations became more volatile

starting in the 1980s and declined substantially in the 2000s (Delaney and Doyle 2011). Public flagship universities responded by enrolling more nonresident students (Jaquette and Curs 2015), who pay higher tuition prices and tend to be more affluent than resident students (Jaquette et al. 2016). Therefore, state disinvestment makes public research universities beholden to the enrollment economy that governs selective private research universities, pursuing national recognition rankings and/or intercollegiate athletics as a means of generating demand from affluent households willing to pay the higher price of nonresident tuition.

Our first research question asks, to what extent are visit patterns consistent with recommendations from the Market Segment Model? The Market Segment Model is an algorithm for identifying desirable recruiting territories. For selective private – and most public – institutions in our sample, the Market Segment Model recommends focusing on affluent Geomarkets located in regions where many students are likely to consider a college like yours (Zemsky and Oedel 1983).

The US postsecondary education system is highly ordinal and so are student market segments. According to Zemsky and Oedel (1983), selective private colleges primarily depend on students from the regional and national market segments. Further, regional and national students tend to come from affluent, college-educated households. Affluent households are not scattered randomly across the country, but rather clump together in particular geographic communities. Like UK “league tables” (McArthur and Reeves 2022), Geomarkets are a means of commensurating geographically disparate localities in terms of the number of students from each market segment. Imagine that the University of Arkansas has substantial regional but not national student demand and wants to enroll more out-of-state students. The Market Segment Model points to the affluent “TX 23 – Collin & Rockwall” (near Dallas) and the “TX 15 – Northwest Houston & Conroe School District” Geomarkets, which contain large numbers of regional and national students, but not the poorer “TX 19 – City of Dallas” or “TX 17 – City of Houston (East),” which contain relatively few regional and national students. We answer RQ1 using simple descriptive statistics. We predict most colleges in our sample will adhere to a regional or national (but not an in-state) recruiting strategy, depending on whether they have regional or national enrollment demand. We also predict that colleges will disproportionately visit high schools located in affluent Geomarkets.

RQ2 asks, to what extent do College Board Geomarkets (X) explain which high schools receive

recruiting visits (Y)? We argue that Geomarkets explain college recruiting visits to high schools because they have become embedded in the organizational and technological infrastructures that generate them. These embeddings exemplify agencement, whereby Geomarkets transform from a mere classificatory device into a durable “socio-technical arrangement” of software, data fields, reports, and organizational roles, making Geomarkets a default unit for planning, staffing, and evaluation of recruitment. Through this agencement, Geomarkets may have performative consequences for recruiting behavior – even after controlling for school- and neighborhood-level covariates – because once a classification scheme is built into workflows, behavior will tend to align with it. We provide three examples of embedding Geomarkets into recruiting infrastructure.

First, although Geomarkets were initially designed to approximate the territories of admissions officers (Zemsky and Oedel 1983), over time they became an organizing principle through which admissions offices define recruiting territories and assign them to officers. On background, admissions officers and EM consultants told us that Geomarkets became part of the shorthand lexicon of the National Association of College Admissions Officers (NACAC). Admissions officers might introduce themselves by saying, “I recruit ‘PA 2’,” referring to the “Chester County, PA” Geomarket. We find evidence for this claim in the LinkedIn profiles of admissions officers. Examples include: “Personally managed schools in Indiana geomarkets 1, 2, 3, 4, 5, 6, 7, & 9” [CITE]; “Led recruitment efforts in geomarket MA06, the greater metro Boston area” [CITE]; and “Increased applications from the Long Island geomarket NY 16 - NY 21 for the 2021 admission cycle by 18.3%” [CITE]. Institutional materials also incorporate Geomarkets; a report from the Dartmouth College Admissions Ambassador Program lists the number of interviewers and applicants by Geomarket [CITE]. Anecdotal evidence suggests that admissions offices allocate relatively more admissions officers to affluent Geomarkets.⁴

Second, similar to the Black-Scholes model being written into spreadsheets used by options traders (MacKenzie and Millo 2003), Geomarkets and the logic of the Market Segment Model became the basis of the widely used EPS software, which encouraged colleges to identify promising Geomarkets and then select high schools to visit within them. EPS promotional material from College Board (2022) includes a screenshot of the user-interface navigation menu, which includes tabs for “Plan Travel,”

⁴For example, the University of Chicago assigns one recruiter to the “IL 9 - North Shore” Geomarket, which had a 2020 population of about 161,000, and one recruiter to the “IL 12 - South and Southwest Suburbs” Geomarket, which had a 2020 population of about 1.2 million [CITE]

“Research High Schools,” and “Competitive Analysis.” This agencement integrates Geomarkets and the Market Segment Model logic into software-based workflows of territory assignment and travel planning, shaping the menu of possible recruiting visits and making some visits more likely than others. On background, EM professionals told us that EPS software enabled colleges to plan travel without relying on admissions officers having strong local knowledge of their territories. EPS software often recommended visiting the same sets of affluent, high-achieving schools visited by other colleges. Some savvier, quantitatively adept offices, however, used EPS to visit schools that EPS recommended ignoring because there would be less competition for the good students who attended these schools.

Third, Geomarkets have been integrated into Customer Relationship Management (CRM) and information systems used by colleges. Slate, the market-leading admissions CRM, allows institutions to assign admissions recruiters and prospective students to Geomarkets [CITE]. The LinkedIn profile of an admissions officer states, “Utilized Slate to create territory management tools for the recruitment team. Generated reports broken down by state and geomarket and sorted by high school to reflect applicant/admit/deposit data for past 3 cycles” [CITE]. Similarly, Oracle’s PeopleSoft Campus Solutions student information system product integrates Geomarkets (referred to as “EPS Market Codes”) and also EPS software [CITE]. These examples illustrate market devices as assemblages of heterogeneous components that build on one another. Integrating Geomarkets within third-party administrative infrastructures makes it easier for colleges to incorporate Geomarkets into recruiting workflows, including decisions about which high schools to visit.

RQ3 asks, to what extent does Geomarket popularity – measured as the number of visits per school – mediate the relationship between school composition and the probability of a receiving a visit? Previous research shows that having a high proportion of free-reduced lunch or Black/Hispanic students is negatively associated with receiving a visit (Salazar et al. 2021). Therefore, RQ3 investigates whether high- versus low-SES schools (and low versus high share of Black/Hispanic enrollment) benefit more from being located in a popular Geomarket. We hypothesize that high-SES and low Black/Hispanic enrollment schools benefit more from being located in a popular Geomarket. Our rationale draws from Fourcade and Healy (2017), who highlight the role of “good matches” between ordinally classified products and ordinally classified customers. Fourcade (2016, 20–21)

offers an anecdote:

One of the authors had a credit card transaction denied while trying to refill her car’s gas tank in a high-poverty Oakland neighborhood. The clerk indicated with resigned anger that the location of his gas station was the likely cause of the problem. Back in Berkeley a few minutes later, the card worked without a hitch.

Not coincidentally, Oakland is in Geomarket “CA 7 – City of Oakland” whereas Berkeley is in Geomarket “CA 8 – Alameda County excluding Oakland.” Fourcade and Healy (2017, 17) write that “‘good matches’ are perceived as natural because they fit well with how things already are.” An affluent school in an affluent Geomarket is a good match because it exemplifies the schools we expect to find. We predict that such schools benefit by receiving more visits than an affluent school in a less affluent Geomarket. By contrast, a poor or predominantly Black/Hispanic school in an affluent Geomarket does not match expectations and will not reap the same locational benefits.

5 Methods

6 Discussion

7 References

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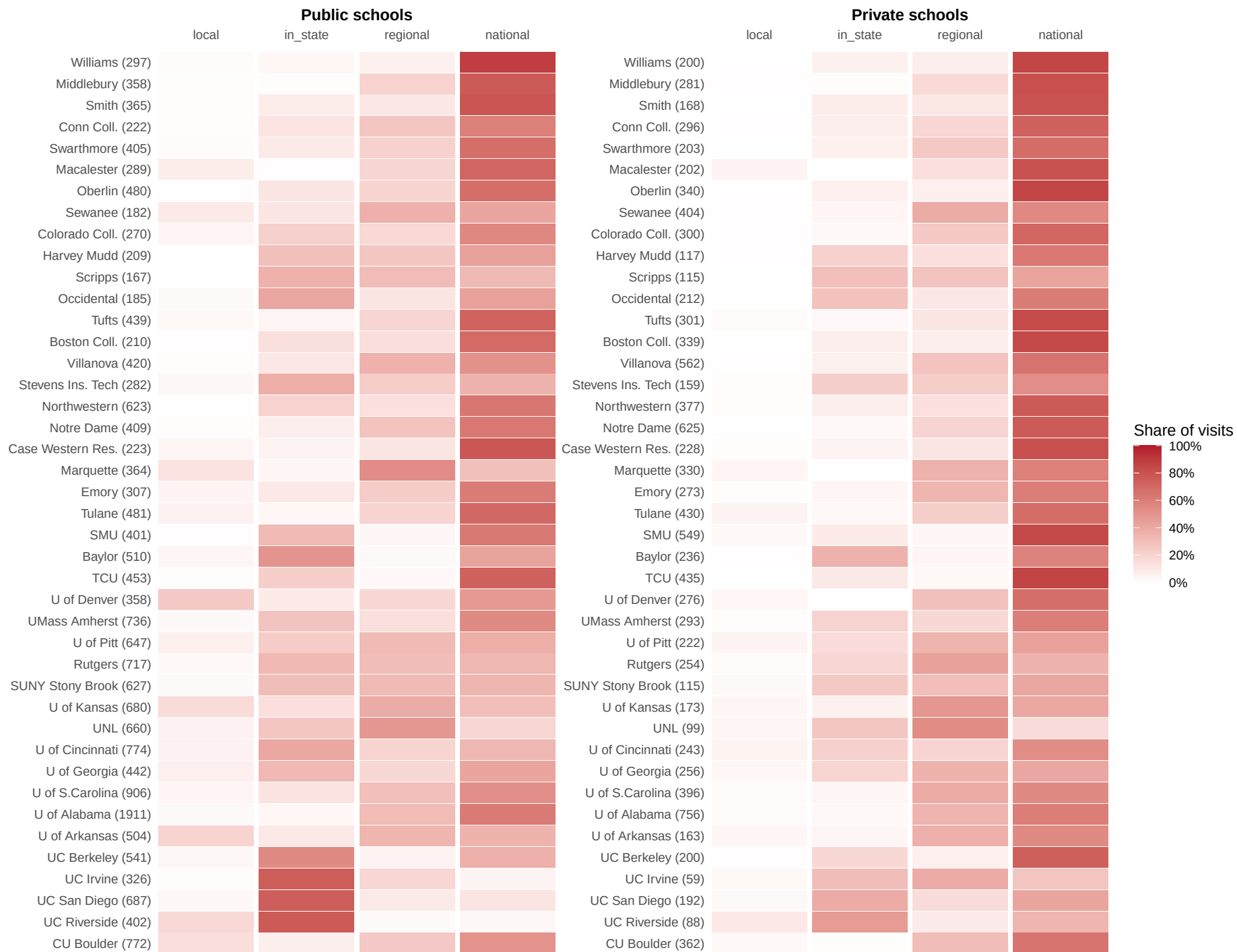


Figure 2: Recruiting visits by market segment of Geomarket

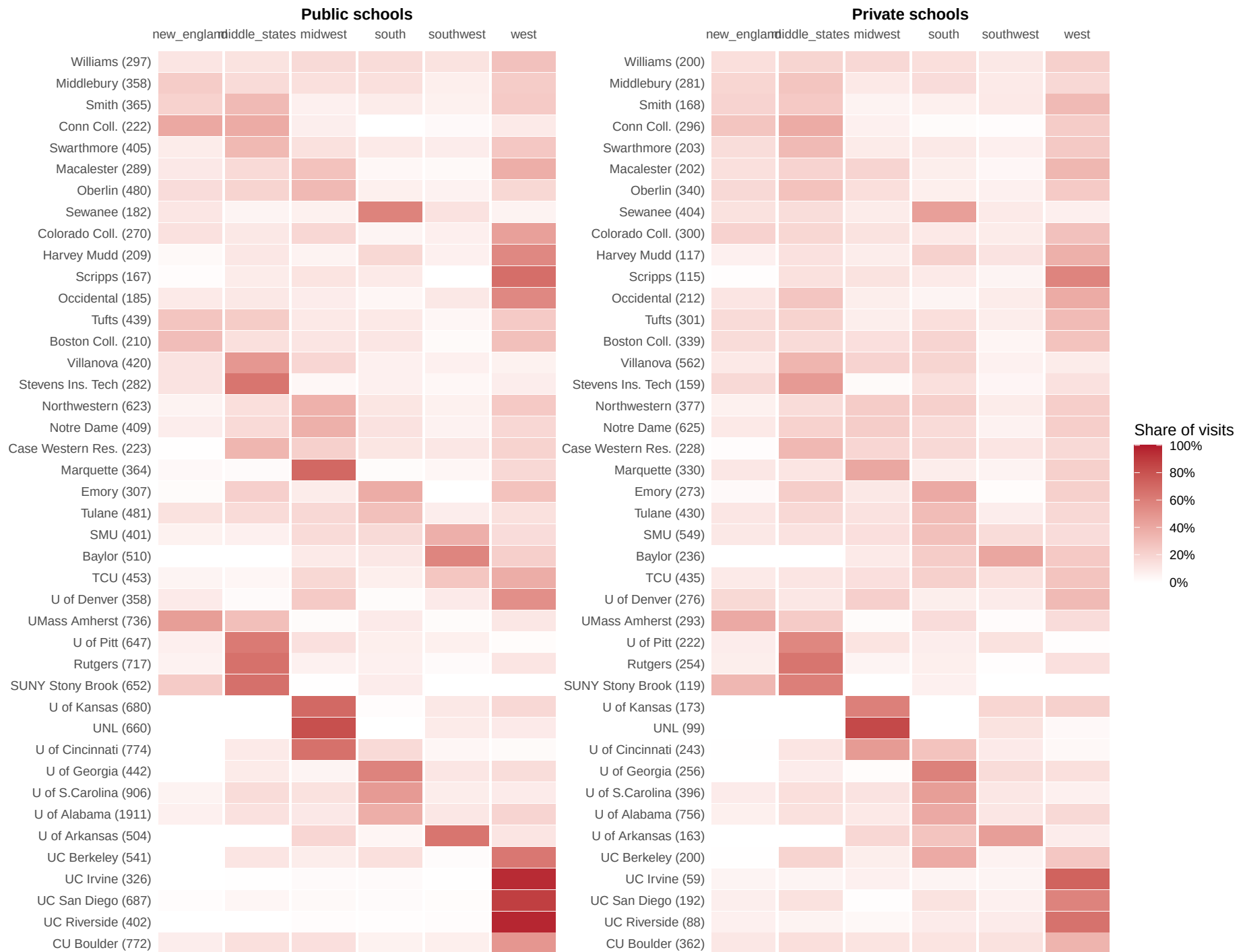


Figure 3: Recruiting visits by EPS region

Table 1: Top 30 Geomarkets ranked by visits per school (public school visits only)

Rank	EPS	Visits/Sch	MeanInc	%BA+	%Pov	%White	%Asian	%Black	%Hisp
1	IL 9 - North Shore	24.5	\$259k	73.5	5.6	79.0	9.6	1.9	7.2
2	IL10 - Evanston & Skokie	17.0	\$156k	58.3	8.4	64.6	17.7	5.1	9.4
3	VA 1 - Arlington & Alexandria	11.6	\$175k	71.4	6.6	57.1	8.8	13.9	15.8
4	TX23 - Collin & Rockwall Co	11.2	\$146k	52.3	6.0	56.8	14.5	9.4	15.6
5	IL 8 - Northwest Suburbs	11.0	\$126k	44.7	6.8	58.2	15.0	3.3	20.8
6	MD 2 - Montgomery Metropolitan	10.4	\$165k	59.2	6.6	43.1	14.9	18.0	19.5
7	VA 2 - Fairfax Co	10.3	\$182k	62.3	5.4	50.4	19.6	9.3	16.2
8	IL12 - Western Suburbs	10.3	\$134k	46.9	6.8	60.3	10.5	8.1	18.5
9	CT 3 - Fairfield Co	9.9	\$160k	48.9	8.7	61.0	5.3	10.6	20.0
10	IL 7 - Chain of Lakes	9.7	\$125k	39.7	8.4	57.2	7.4	7.5	25.1
11	CA11 - Santa Clara Co (excluding San Jose)	9.6	\$234k	64.3	5.7	36.8	39.4	1.8	17.1
12	CO 2 - Metro Denver & Northeastern Colorado	9.4	\$121k	44.8	8.7	66.2	3.8	4.3	22.1
13	CA 4 - Marin Co	9.0	\$190k	60.2	6.6	70.6	5.7	2.1	16.1
14	NY17 - Northern Nassau Co	8.8	\$266k	62.4	6.7	67.7	16.9	2.3	10.6
15	NJ 6 - Somerset & Mercer Co	8.8	\$155k	49.3	8.5	51.8	14.6	14.8	16.4
16	CA28 - South Orange Co	8.2	\$177k	56.7	7.2	57.4	19.5	1.4	16.5
17	CA17 - West Los Angeles & West Beach	8.2	\$155k	60.3	11.1	50.3	11.9	9.6	22.5
18	MA10 - Milton, Lexington, & Waltham	8.1	\$181k	64.0	5.8	74.2	13.1	3.4	6.0
19	GA 2 - Fulton Co	7.7	\$116k	54.5	12.3	39.3	7.3	43.1	7.2
20	NY19 - Northwest Suffolk Co	7.7	\$182k	47.7	4.6	76.9	5.9	3.0	12.0
21	CA 6 - Contra Costa Co	7.2	\$151k	43.3	8.1	42.6	17.2	8.2	25.8
22	CA26 - Santa Ana	7.0	\$131k	36.5	10.4	35.1	21.1	1.1	38.9
23	TX24 - West of Dallas/Ft. Worth Metroplex	6.9	\$124k	41.3	6.9	62.6	6.9	8.9	17.9
24	NJ11 - Morris & Northern Passaic Co	6.8	\$173k	53.4	5.3	71.8	9.7	2.9	13.2
25	NJ10 - Bergen Co	6.7	\$159k	50.7	7.2	55.5	16.4	5.3	20.4
26	CA 8 - Alameda Co (w/o Oakland)	6.7	\$165k	49.3	7.6	31.0	36.4	6.0	20.8
27	NJ 7 - Union Co	6.6	\$114k	37.1	8.9	39.0	5.2	20.2	32.0
28	PA 4 - Montgomery Co	6.6	\$142k	49.6	6.2	74.9	7.7	9.1	5.3
29	CA20 - South Bay	6.5	\$126k	44.6	8.9	31.7	21.8	9.8	31.7
30	MA 8 - Lowell, Concord, & Wellesley	6.5	\$160k	51.6	7.2	75.0	10.7	3.4	7.4

Table 2: Top 30 Geomarkets ranked by visits per school (private school visits only)

Rank	EPS	Visits/Sch	MeanInc	%BA+	%Pov	%White	%Asian	%Black	%Hisp
1	IL 9 - North Shore	31.0	\$259k	73.5	5.6	79.0	9.6	1.9	7.2
2	TX19 - City of Dallas	22.4	\$81k	34.9	15.6	29.8	3.4	22.9	41.7
3	IL10 - Evanston & Skokie	20.3	\$156k	58.3	8.4	64.6	17.7	5.1	9.4
4	TX20 - City of Fort Worth	20.2	\$91k	28.3	12.1	42.0	4.4	15.8	34.8
5	CA 7 - City of Oakland	19.5	\$124k	47.0	13.4	29.6	15.7	21.7	26.3
6	IL 7 - Chain of Lakes	19.0	\$125k	39.7	8.4	57.2	7.4	7.5	25.1
7	VA 1 - Arlington & Alexandria	18.8	\$175k	71.4	6.6	57.1	8.8	13.9	15.8
8	CA 4 - Marin Co	17.7	\$190k	60.2	6.6	70.6	5.7	2.1	16.1
9	DC 1 - District of Columbia	17.0	\$155k	59.8	13.7	36.7	4.0	44.5	11.1
10	TX16 - Southwest Houston Metro Area	16.6	\$106k	46.8	11.8	31.5	16.1	20.0	29.6
11	CA29 - North San Diego Co (w/o San Diego)	16.0	\$144k	44.1	8.0	52.3	10.4	2.3	29.8
12	CT 4 - Waterbury & Litchfield Co	16.0	\$129k	35.9	7.2	87.6	1.9	1.5	6.6
13	NH 1 - Seacoast	16.0	\$138k	44.5	7.1	91.2	3.0	0.9	2.3
14	GA 2 - Fulton Co	14.9	\$116k	54.5	12.3	39.3	7.3	43.1	7.2
15	CA28 - South Orange Co	14.2	\$177k	56.7	7.2	57.4	19.5	1.4	16.5
16	CT 5 - Hartford & Tolland Co	14.1	\$117k	39.0	11.0	63.5	5.4	11.4	16.6
17	CA31 - City of San Diego	14.0	\$118k	45.4	11.4	44.7	14.7	6.8	28.9
18	MA 7 - Quincy & Plymouth Co	13.2	\$137k	38.6	7.5	80.2	1.4	8.9	4.0
19	TX 6 - Austin & Central Texas	13.1	\$114k	43.0	9.4	53.7	5.0	6.4	31.9
20	NY19 - Northwest Suffolk Co	13.0	\$182k	47.7	4.6	76.9	5.9	3.0	12.0
21	OH 5 - Cuyahoga, Geauga, & Lake Co	12.8	\$106k	36.4	7.1	87.2	2.3	4.6	3.5
22	OR 1 - Greater Portland (West)	12.8	\$130k	53.6	9.4	69.6	9.0	3.3	12.1
23	MO 1 - Kansas City & St. Joseph	12.6	\$84k	28.7	11.9	76.0	1.5	11.9	6.7
24	NJ 6 - Somerset & Mercer Co	12.2	\$155k	49.3	8.5	51.8	14.6	14.8	16.4
25	CA15 - San Fernando Valley (East)	12.1	\$123k	51.6	12.0	58.4	6.8	5.1	24.9
26	CO 2 - Metro Denver & Northeastern Colorado	12.0	\$121k	44.8	8.7	66.2	3.8	4.3	22.1
27	MD 2 - Montgomery Metropolitan	11.9	\$165k	59.2	6.6	43.1	14.9	18.0	19.5
28	CA11 - Santa Clara Co (excluding San Jose)	11.6	\$234k	64.3	5.7	36.8	39.4	1.8	17.1
29	CA26 - Santa Ana	11.5	\$131k	36.5	10.4	35.1	21.1	1.1	38.9
30	CT 2 - New Haven & Middlesex Co	11.3	\$118k	37.1	10.8	65.3	3.9	11.3	16.6

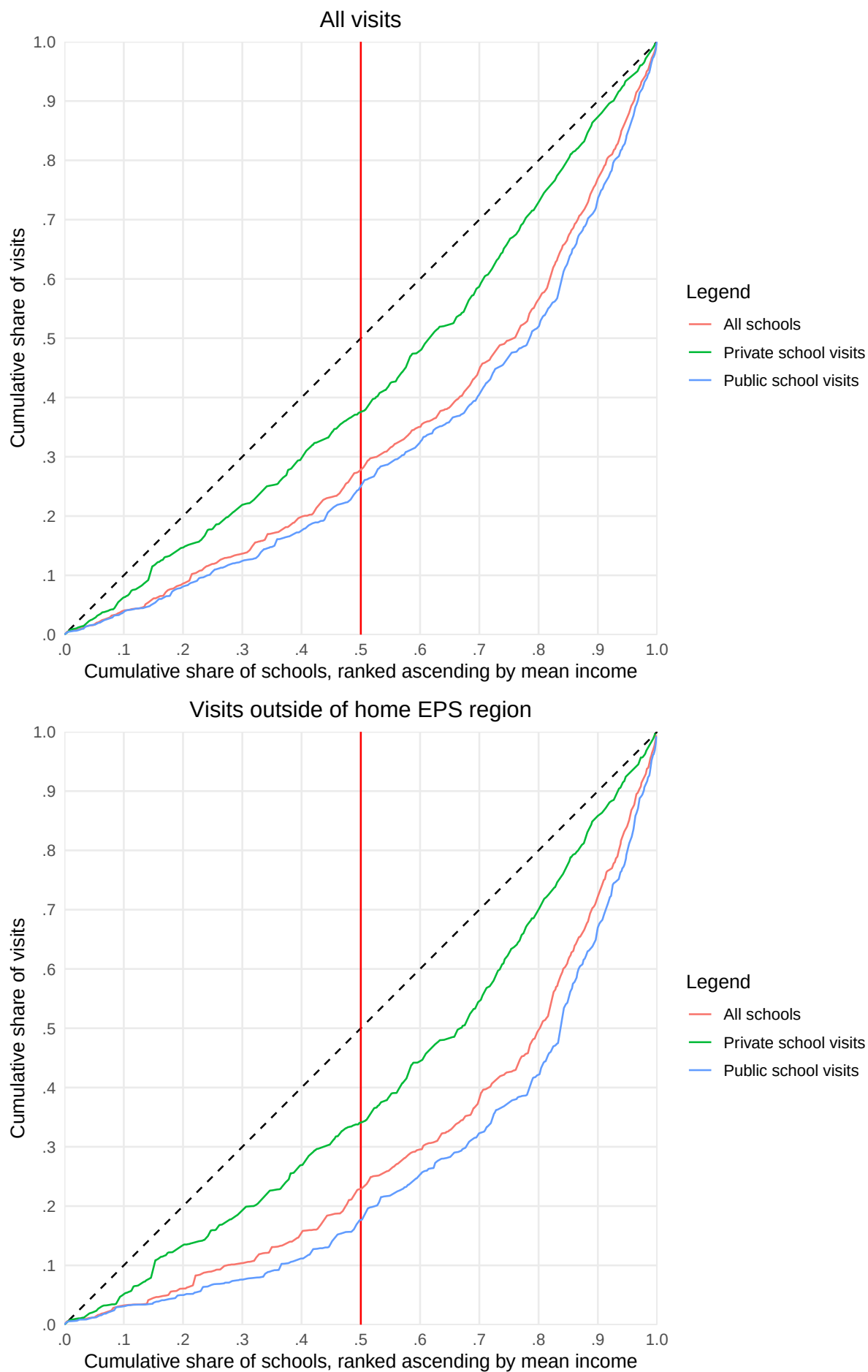


Figure 4: Lorenz-style concentration curves of recruiting visits by Geomarket affluence

Table 3: Model Fit Statistics, fixed effects regression of visit to high school i by college j, separate models for all HS, public HS, private HS

term	(1) Univ $\times$ State FE	(2) Univ $\times$ EPS FE	(3) Covar + Univ FE	(4) Covar + Univ $\times$ State FE	(5) Covar + Univ $\times$ EPS FE
All High Schools					
N	1,026,984	1,026,984	1,013,166	1,013,166	1,013,166
Adj. R ²	0.097	0.168	0.136	0.210	0.258
Within R ²	0.000	0.000	0.129	0.125	0.108
RMSE	0.165	0.158	0.162	0.155	0.149
Public High Schools Only					
N	853,692	853,692	748,440	748,440	748,440
Adj. R ²	0.118	0.210	0.120	0.223	0.301
Within R ²	0.000	0.000	0.111	0.104	0.082
RMSE	0.143	0.134	0.149	0.140	0.132
Private High Schools Only					
N	173,292	173,292	167,454	167,454	167,454
Adj. R ²	0.096	0.146	0.170	0.229	0.264
Within R ²	0.000	0.000	0.153	0.147	0.137
RMSE	0.238	0.224	0.232	0.222	0.210

Table 4: Fixed Effects Regression Model with Interaction between Geomarket Popularity and School SES/Racial Composition

	Model		Interaction Free-Reduced Lunch		Interaction Black/Hispanic	
	Estimate	Std Error	Estimate	Std Error	Estimate	Std Error
hs_pct_free_reduced_lunch_decileD2	-0.008*	0.003	0.016***	0.003	-0.007*	0.003
hs_pct_free_reduced_lunch_decileD3	-0.006	0.004	0.025***	0.004	-0.004	0.003
hs_pct_free_reduced_lunch_decileD4	-0.005	0.004	0.030***	0.004	-0.003	0.004
hs_pct_free_reduced_lunch_decileD5	-0.003	0.004	0.034***	0.004	-0.001	0.004
hs_pct_free_reduced_lunch_decileD6	-0.004	0.004	0.033***	0.004	-0.001	0.004
hs_pct_free_reduced_lunch_decileD7	-0.004	0.004	0.035***	0.004	-0.001	0.004
hs_pct_free_reduced_lunch_decileD8	-0.002	0.004	0.036***	0.005	0.001	0.004
hs_pct_free_reduced_lunch_decileD9	-0.001	0.005	0.038***	0.005	0.002	0.004
hs_pct_free_reduced_lunch_decileD10	-0.000	0.004	0.043***	0.004	0.002	0.004
hs_pct_bl_hisp_nat_decileD1	0.005	0.002	0.006*	0.002	0.020***	0.003
hs_pct_bl_hisp_nat_decileD2	0.003	0.002	0.004	0.002	0.012***	0.003
hs_pct_bl_hisp_nat_decileD3	0.003	0.003	0.003	0.002	0.006*	0.002
hs_pct_bl_hisp_nat_decileD4	0.002	0.002	0.001	0.002	0.002	0.002
hs_pct_bl_hisp_nat_decileD6	-0.005***	0.001	-0.004**	0.001	-0.002	0.002
hs_pct_bl_hisp_nat_decileD7	-0.004*	0.002	-0.001	0.002	0.001	0.003
hs_pct_bl_hisp_nat_decileD8	-0.007*	0.003	-0.002	0.002	0.004	0.003
hs_pct_bl_hisp_nat_decileD9	-0.011*	0.004	-0.003	0.004	0.006	0.003
hs_pct_bl_hisp_nat_decileD10	-0.016**	0.005	-0.005	0.005	0.012**	0.004
peps_n_vis01_per_sch_all	0.504***	0.035	0.955***	0.049	0.695***	0.046
hs_pct_free_reduced_lunch_decileD2:peps_n_vis01_per_sch_all			-0.327***	0.032		
hs_pct_free_reduced_lunch_decileD3:peps_n_vis01_per_sch_all			-0.480***	0.041		
hs_pct_free_reduced_lunch_decileD4:peps_n_vis01_per_sch_all			-0.569***	0.047		
hs_pct_free_reduced_lunch_decileD5:peps_n_vis01_per_sch_all			-0.629***	0.048		
hs_pct_free_reduced_lunch_decileD6:peps_n_vis01_per_sch_all			-0.609***	0.047		
hs_pct_free_reduced_lunch_decileD7:peps_n_vis01_per_sch_all			-0.675***	0.043		
hs_pct_free_reduced_lunch_decileD8:peps_n_vis01_per_sch_all			-0.665***	0.059		
hs_pct_free_reduced_lunch_decileD9:peps_n_vis01_per_sch_all			-0.697***	0.071		
hs_pct_free_reduced_lunch_decileD10:peps_n_vis01_per_sch_all			-0.813***	0.039		
hs_pct_bl_hisp_nat_decileD1:peps_n_vis01_per_sch_all					-0.456***	0.067
hs_pct_bl_hisp_nat_decileD2:peps_n_vis01_per_sch_all					-0.200**	0.058
hs_pct_bl_hisp_nat_decileD3:peps_n_vis01_per_sch_all					-0.067	0.059
hs_pct_bl_hisp_nat_decileD4:peps_n_vis01_per_sch_all					-0.005	0.040
hs_pct_bl_hisp_nat_decileD6:peps_n_vis01_per_sch_all					-0.089**	0.028
hs_pct_bl_hisp_nat_decileD7:peps_n_vis01_per_sch_all					-0.128***	0.035
hs_pct_bl_hisp_nat_decileD8:peps_n_vis01_per_sch_all					-0.223***	0.042
hs_pct_bl_hisp_nat_decileD9:peps_n_vis01_per_sch_all					-0.316***	0.052
hs_pct_bl_hisp_nat_decileD10:peps_n_vis01_per_sch_all					-0.487***	0.058
N	678,804		678,804		678,804	
Adj. R²	0.272		0.298		0.282	
Within R²	0.151		0.181		0.162	
RMSE	0.140		0.138		0.139	

Significance codes: *** p<0.001, ** p<0.01, * p<0.05

8 Appendix

Table A1: Fixed effects regression models of visit to high school i by college j (public high schools only)

term	(1) Univ $\times$ State FE	(2) Univ $\times$ EPS FE	(3) Covar + Univ FE	(4) Covar + Univ $\times$ State FE	(5) Covar + Univ $\times$ EPS FE
hs_g11			0.000*** (0.000)	0.000*** (0.000)	0.000*** (0.000)
hs_pct_asian			0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
hs_pct_black			-0.000*** (0.000)	-0.000* (0.000)	-0.000 (0.000)
hs_pct_hispanic			-0.000 (0.000)	-0.000* (0.000)	-0.000** (0.000)
hs_pct_amerindian			-0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
hs_pct_nativehawaii			0.001** (0.000)	-0.001** (0.000)	-0.001*** (0.000)
hs_pct_tworaces			0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
hs_overall_niche_letter_gradeA+			0.126*** (0.013)	0.128*** (0.013)	0.120*** (0.011)
hs_overall_niche_letter_gradeA			0.027*** (0.006)	0.028*** (0.005)	0.027*** (0.005)
hs_overall_niche_letter_gradeA-			0.004 (0.003)	0.005 (0.002)	0.005 (0.003)
hs_overall_niche_letter_gradeB+			0.000 (0.002)	-0.000 (0.002)	0.000 (0.002)
hs_overall_niche_letter_gradeB			-0.002 (0.001)	-0.002 (0.002)	-0.002 (0.002)
hs_overall_niche_letter_gradeB-			-0.004** (0.001)	-0.004* (0.002)	-0.003* (0.002)
hs_overall_niche_letter_gradeC+			-0.004** (0.001)	-0.004* (0.002)	-0.003 (0.002)
hs_overall_niche_letter_gradeC			-0.003* (0.001)	-0.004* (0.001)	-0.003 (0.002)
hs_overall_niche_letter_gradeC-			-0.003* (0.001)	-0.005** (0.001)	-0.004* (0.002)
hs_magnet011			-0.006 (0.003)	-0.002 (0.003)	-0.000 (0.003)
hs_school_typecareer and technical school			-0.029** (0.009)	-0.034** (0.010)	-0.030** (0.009)
hs_school_typealternative education school			-0.005* (0.002)	0.003 (0.003)	0.004 (0.003)
hs_pct_free_reduced_lunch			-0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)

(continued)

term	(1) Univ $\times$ State FE	(2) Univ $\times$ EPS FE	(3) Covar + Univ FE	(4) Covar + Univ $\times$ State FE	(5) Covar + Univ $\times$ EPS FE
hs_pct_prof_math			-0.020** (0.007)	0.002 (0.006)	0.004 (0.005)
hs_pct_prof_rla			0.006 (0.007)	0.000 (0.005)	0.001 (0.005)
hs_zip_inc_house_mean_decileD2			0.002 (0.001)	0.000 (0.001)	0.000 (0.001)
hs_zip_inc_house_mean_decileD3			0.002 (0.002)	-0.001 (0.001)	-0.001 (0.001)
hs_zip_inc_house_mean_decileD4			0.003 (0.002)	0.000 (0.002)	0.000 (0.002)
hs_zip_inc_house_mean_decileD5			0.003 (0.003)	-0.001 (0.002)	-0.001 (0.002)
hs_zip_inc_house_mean_decileD6			0.004 (0.003)	-0.002 (0.002)	-0.001 (0.002)
hs_zip_inc_house_mean_decileD7			0.005 (0.003)	-0.001 (0.002)	-0.002 (0.002)
hs_zip_inc_house_mean_decileD8			0.007 (0.004)	-0.003 (0.002)	-0.004 (0.002)
hs_zip_inc_house_mean_decileD9			0.011* (0.005)	-0.000 (0.003)	-0.002 (0.003)
hs_zip_inc_house_mean_decileD10			0.040*** (0.009)	0.027*** (0.006)	0.023** (0.007)
hs_zip_pct_edu_baplus_all_decileD2			-0.000 (0.001)	0.000 (0.001)	-0.001 (0.001)
hs_zip_pct_edu_baplus_all_decileD3			-0.001 (0.001)	-0.001 (0.001)	-0.002** (0.001)
hs_zip_pct_edu_baplus_all_decileD4			-0.000 (0.001)	-0.001 (0.001)	-0.003* (0.001)
hs_zip_pct_edu_baplus_all_decileD5			0.001 (0.001)	-0.000 (0.002)	-0.002 (0.001)
hs_zip_pct_edu_baplus_all_decileD6			-0.001 (0.002)	-0.002 (0.002)	-0.004* (0.001)
hs_zip_pct_edu_baplus_all_decileD7			0.001 (0.002)	-0.000 (0.002)	-0.003 (0.002)
hs_zip_pct_edu_baplus_all_decileD8			0.001 (0.003)	0.001 (0.002)	-0.003 (0.002)
hs_zip_pct_edu_baplus_all_decileD9			0.009 (0.005)	0.008** (0.003)	0.004 (0.003)
hs_zip_pct_edu_baplus_all_decileD10			0.049*** (0.009)	0.048*** (0.008)	0.041*** (0.007)
hs_zip_pct_pov_yes			0.000	0.000	0.000

(continued)

term	(1) Univ × State FE	(2) Univ × EPS FE	(3) Covar + Univ FE	(4) Covar + Univ × State FE	(5) Covar + Univ × EPS FE
			(0.000)	(0.000)	(0.000)
I(hs_zip_pct_pov_yes^2)			0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
hs_zip_pct_nhisp_black			0.000*** (0.000)	0.000*** (0.000)	0.000 (0.000)
hs_zip_pct_nhisp_native			0.000 (0.000)	0.000 (0.000)	0.000 (0.000)
hs_zip_pct_nhisp_asian			0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)
hs_zip_pct_nhisp_nhpi			0.001 (0.000)	0.000 (0.000)	-0.000 (0.000)
hs_zip_pct_nhisp_multi			0.001* (0.000)	-0.000 (0.000)	-0.000 (0.000)
hs_zip_pct_hisp_all			0.000* (0.000)	0.000 (0.000)	0.000 (0.000)
hs_univ_dist			-0.000*** (0.000)	-0.000** (0.000)	-0.000* (0.000)
N	853,692	853,692	748,440	748,440	748,440
Adj. R ²	0.118	0.210	0.120	0.223	0.301
Within R ²	0.000	0.000	0.111	0.104	0.082
RMSE	0.143	0.134	0.149	0.140	0.132

Significance codes: *** p<0.001, ** p<0.01, * p<0.05

Table A2: Model fit statistics, fixed effects regression models of visits to high school i, separate models models by college j (public high schools only)

University	USNWR Rank	# PubHS Visits	AdjR2 State Only	AdjR2 EPS FE Only	AdjR2 Cov Only	AdjR2 Cov State	AdjR2 Cov EPS FE
Private Liberal Arts							
Williams	1	305	0.00	0.04	0.10	0.11	0.13
Swarthmore	3	441	0.02	0.09	0.10	0.12	0.17
Middlebury	9	376	0.03	0.08	0.12	0.14	0.18
Smith	15	404	0.03	0.10	0.11	0.12	0.17
Harvey Mudd	25	216	0.01	0.04	0.09	0.09	0.12
Colorado Coll.	25	300	0.05	0.09	0.12	0.16	0.19
Macalester	27	311	0.02	0.06	0.11	0.12	0.15
Scripps	28	169	0.02	0.06	0.07	0.08	0.13
Oberlin	36	502	0.02	0.09	0.20	0.22	0.25
Occidental	40	193	0.02	0.06	0.09	0.10	0.16
Sewanee	47	206	0.03	0.07	0.06	0.09	0.13
Conn Coll.	51	225	0.05	0.12	0.11	0.15	0.20
Private Research							
Northwestern	9	640	0.03	0.13	0.22	0.25	0.31
Notre Dame	19	422	0.01	0.07	0.16	0.18	0.21
Emory	21	315	0.02	0.11	0.13	0.15	0.22
Tufts	30	462	0.04	0.12	0.14	0.17	0.23
Boston Coll.	35	215	0.02	0.06	0.08	0.10	0.13
Tulane	41	503	0.03	0.13	0.17	0.20	0.26
Case Western Res.	42	227	0.01	0.06	0.11	0.12	0.16
Villanova	53	455	0.06	0.14	0.16	0.19	0.24
SMU	66	445	0.02	0.08	0.14	0.16	0.19
Baylor	76	580	0.08	0.15	0.14	0.20	0.24
U of Denver	80	413	0.11	0.16	0.12	0.21	0.26
TCU	80	509	0.02	0.11	0.16	0.18	0.24
Stevens Ins. Tech	80	310	0.12	0.20	0.09	0.17	0.24
Marquette	88	424	0.07	0.19	0.10	0.18	0.27
Public Research							
UC Berkeley	22	595	0.08	0.17	0.17	0.21	0.32
UC Irvine	35	436	0.09	0.16	0.12	0.17	0.25
UC San Diego	35	792	0.18	0.26	0.26	0.36	0.45
U of Georgia	47	514	0.16	0.22	0.11	0.25	0.30
U of Pitt	58	809	0.12	0.26	0.18	0.27	0.36
Rutgers	63	1065	0.25	0.40	0.22	0.37	0.48
UMass Amherst	66	768	0.24	0.35	0.19	0.37	0.43
UC Riverside	88	529	0.15	0.33	0.19	0.29	0.51
SUNY Stony Brook	88	842	0.19	0.45	0.19	0.29	0.49
CU Boulder	103	882	0.11	0.21	0.27	0.36	0.40
U of S.Carolina	118	1224	0.16	0.26	0.22	0.35	0.40
U of Kansas	124	966	0.24	0.31	0.15	0.36	0.41
UNL	133	1128	0.33	0.39	0.10	0.39	0.45
U of Alabama	143	2594	0.08	0.18	0.26	0.32	0.36
U of Cincinnati	143	926	0.22	0.36	0.12	0.31	0.42
U of Arkansas	160	731	0.19	0.25	0.09	0.26	0.31